SmartAssist: AI-Powered Triage and Specialist Appointment Booking System

Project Documentation

Team Name:

Team 3: Triage and Specialist Appointment Booking System

Team Members:

- Aditya
- Atharsh

Company: Ullavi Technologies

Duration: May 10, 2026 to June 10, 2026

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Executive Summary

SmartAssist is a state-of-the-art AI-powered triage and specialist appointment booking platform designed to streamline patient-doctor interactions. Developed with a high-performance frontend using React 19 and Vite, and a backend powered by Node.js, Express, and MongoDB, SmartAssist simplifies clinical triage and scheduling. By integrating conversational artificial intelligence with real-time specialist availability, the application provides users with immediate, intelligent guidance concerning their health concerns while optimizing administrative workflows for medical practitioners.

At the core of SmartAssist is an interactive, three-turn AI Triage Nurse chat system. Users enter their details and describe their symptoms either through text or via voice recording using the MediaRecorder API. The recorded audio is processed by the backend transcription service and analyzed to extract key symptoms and match confidence levels. The system then recommends the most suitable specialist, provides an urgency level, and displays emergency warnings for critical conditions, ensuring patients receive timely care.

SmartAssist supports a structured specialist management ecosystem. Healthcare professionals register via a detailed verification process that requires admin approval. Once approved, specialists manage their profiles, set custom pricing and durations for In-Person, Video Call, or Chat Consultations, and configure availability slots. The booking engine performs real-time slot conflict detection against MongoDB, and includes a automated waitlist management system that allows users to hold and claim newly vacated appointment slots.

The platform offers specialized dashboards for patients and healthcare providers. Patients use their dashboard to track bookings, manage family profiles, join waitlists, and complete post-visit check-ins. An interactive SVG recovery timeline visualizes symptom progression over time, while smart rebooking prompts encourage follow-up care. Meanwhile, the specialist dashboard provides upcoming booking tables, comprehensive pre-visit summaries containing patient triage history, profile update screens, and detailed earnings reports.

To ensure a seamless user experience, SmartAssist integrates background cron services for automated notifications and document delivery. Patients and specialists receive multi-channel reminders via email and SMS at set pre- and post-visit intervals. Additionally, the system generates premium, highly stylized appointment receipts and clinical pre-visit summaries as PDFs. These are emailed directly to specialists before appointments, ensuring they are fully briefed on patient context prior to the consultation.

Problem Statement

Existing Challenges

- **Inefficient Clinical Triage:** Traditional triage processes rely heavily on manual assessment by healthcare staff, leading to administrative bottlenecks, human error in routing, and increased wait times for patients seeking immediate clinical direction.
- **Difficulty in Specialist Matching:** Patients frequently struggle to identify the correct specialist for their specific medical concerns. This often leads to misrouted bookings, delayed treatments, and unnecessary visits to incorrect clinics.
- **Rigid Scheduling and Open Slots:** Conventional appointment systems fail to handle cancellations or schedule adjustments dynamically. When appointments are cancelled, slots often remain unoccupied, leaving healthcare providers underutilized and waitlisted patients without care.
- **Lack of Pre-Visit Context:** Specialists often begin consultations without any structured background on the patient's symptoms or triage history. This forces them to spend valuable session time on basic intake questions rather than active clinical consultation.

Need for Solution

There is an urgent need for an automated, intelligent platform that streamlines symptom triage, matches patients with appropriate specialists, and automates scheduling to reduce wait times and improve clinical workflows.

Model Solution

- Provide a conversational, multi-turn AI chat interface for users to describe symptoms in plain language.
- Implement voice recognition capabilities to allow hands-free triage description and automated transcription.
- Apply real-time classification to route patients to the correct specialist category based on symptom analysis.
- Manage scheduling dynamically, allowing real-time booking, automated waitlists, and slot reallocation on cancellation.
- Generate and deliver digital receipts and structured pre-visit clinical summaries to doctors and patients automatically.

Proposed Solution

Our proposed solution, SmartAssist, addresses these challenges by delivering an end-to-end web-based platform that combines an AI-powered triage nurse with a live booking system. Patients can describe their issues via text or voice, receive immediate specialist recommendations and urgency assessments, and book appointments directly. SmartAssist eliminates the bottlenecks of traditional scheduling through automated slot allocation, a dynamic waitlist system, and instant pre-visit clinical reports, significantly improving both patient access and provider efficiency.

Project Objective

The primary objective of the SmartAssist project is to design and implement an end-to-end, AI-powered health triage and appointment scheduling application that bridges the communication gap between patients and specialized healthcare practitioners. By leveraging advanced natural language understanding and automated scheduling workflows, the system aims to minimize clinic booking bottlenecks, reduce human errors in patient routing, and maximize clinical utilization rates. Through an intuitive, accessible user interface and robust backend synchronization, SmartAssist empowers patients to receive immediate, intelligent symptom feedback and book matching specialist consultations seamlessly.

Specific Objective

- Deliver a conversational, multi-turn AI triage nurse interface for symptom assessment.
- Enable hands-free audio recording and automated speech-to-text transcription for symptom input.
- Perform real-time symptom classification to recommend appropriate medical or wellness specialist categories.
- Provide a dynamic specialist directory with name-based search, bios, ratings, and filters.
- Establish a secure, JWT-based user authentication and registration system with password encryption.
- Support multi-modal appointment options:
    - In-Person consultations at physical clinics with standard pricing structures.
    - Video Call sessions to enable remote clinical evaluations.
    - Chat Consultations for low-urgency questions and quick advice.
- Synchronize real-time appointment bookings to avoid scheduling conflicts across MongoDB collections.
- Deploy an automated waitlist management system to claim newly vacated appointment slots.
- Compile patient symptoms, chat history, and urgency levels into downloadable pre-visit PDF summaries.

Scope of the Project

The frontend scope of SmartAssist encompasses the complete user interface and client-side application logic designed for patients, specialists, and administrators. Built with React 19 and Vite, the frontend delivers a conversational triage chat interface supporting text and voice recording via the MediaRecorder API. It includes a responsive specialist directory with dynamic search and category filters, an interactive calendar for slot booking, and security-guarded user dashboards. Additionally, the client interface provides features like jsPDF-generated past appointment receipts, interactive SVG-based recovery progression timelines, and responsive mobile-first navigation.

The backend scope centers on a Node.js and Express.js server integrated with a MongoDB Atlas database for persistent data storage. It handles JWT-based user authentication, role-based access control middleware, and onboarding workflows for medical specialists. The backend integrates third-party AI APIs (including Gemini, OpenAI, and Groq) to analyze triage discussions and classify symptom urgency, falling back to local keyword-based logic when necessary. It also supports file upload handling for audio-to-text transcriptions and hosts admin tools to manage bookings and specialist approvals.

The background service scope covers real-time waitlist control and multi-channel user notification systems. When an appointment slot is fully booked, the backend waitlist module allows registered patients to join a queue. A background job executes every minute to check and release expired slot holds. If an appointment is cancelled, the system automatically reallocates the slot to the oldest pending waitlist entry. The scope also includes automated pre-appointment reminders at 24h, 2h, and 15m intervals, alongside post-visit symptom check-ins delivered via email and SMS.

Finally, the scope details custom onboarding flows and tools for approved practitioners. New specialists register with license documentation, and applications are held in a pending state until approved or rejected by an admin. Once verified, specialists gain dashboard access to modify their active appointment modes (In-Person, Video, or Chat), adjust mode-specific pricing, and define recurring availability slots. The backend automatically compiles and emails pre-visit PDF summaries containing patient triage history, symptom categories, and urgency levels to practitioners one hour before each consultation.

Team Members, Roles and Responsibilities

Team Member	Roles and Responsibilities
Aaditya	• Designed and built React 19 + Vite frontend using custom CSS. • Developed Node.js + Express backend RESTful API endpoints and CORS routing. • Integrated MongoDB Atlas collections and models (User, Specialist, Booking). • Implemented specialist directory with search, ratings, and filters. • Developed real-time calendar booking integrated with MongoDB. • Created automated waitlist queue, claim holds, and slot reallocator. • Designed user dashboards with family profiles, recovery charts, and symptom check-ins. • Built specialist dashboard with slots, pricing, and earnings management. • Built admin console for bookings and specialist onboarding reviews. • Setup JWT-based authentication and bcrypt password hashing. • Coded plain-language error translator utility and FAQ support chat widget. • Setup MediaRecorder API voice recording and audio transcription interface. • Integrated Google Maps API for clinic location tracking. • Configured Redis caching system for faster server response times. • Implemented Groq AI model integration for fallback recommendation. • Setup cron background tasks and Resend email/SMS dispatches.

	• Deployed apps on Vercel, Hugging Face, and Northflank with load balancing. • Created Docker deployment configurations. • Integrated client jsPDF and server-side clinical summary PDF builders.
Adarsh	• Built conversational multi-turn AI triage nurse interface. • Configured Gemini and OpenAI API integration for symptom triage. • Seeded database collections with mock specialist and triage data. • Integrated Firebase authentication databases and server sync. • Integrated i18n package for multi-language selection support.

Technologies Used

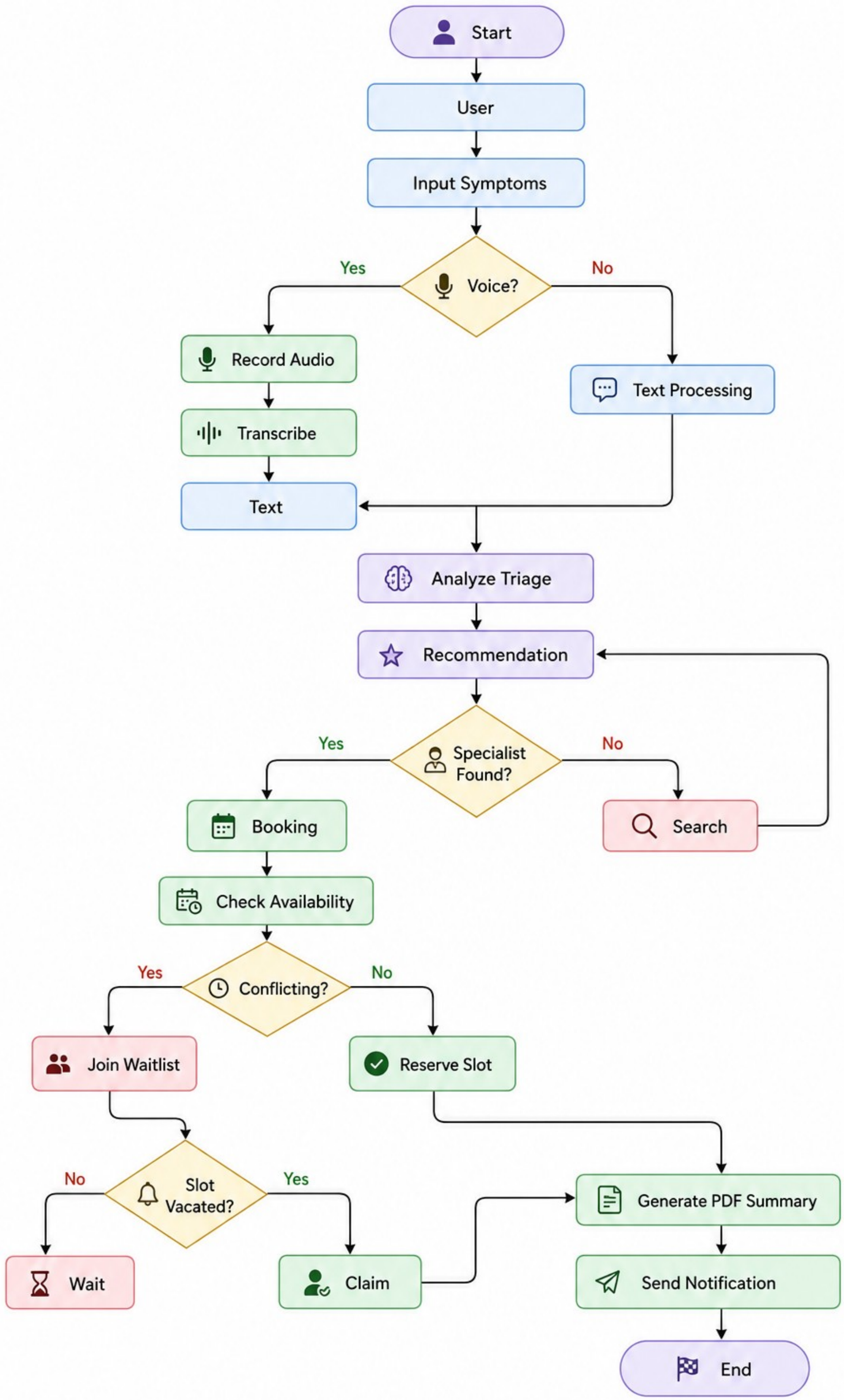
The SmartAssist Platform was developed using a modern, full-stack architecture designed to ensure high performance, scalability, and security. By integrating advanced frontend frameworks, robust backend service layers, and artificial intelligence APIs, the system delivers a seamless healthcare management experience. The following sections detail the core technologies utilized throughout the development of the platform.

Technology	Purpose in the Project
React 19	Client-side library for building dynamic, responsive user and specialist interfaces.
Vite	Next-generation frontend tooling and bundler for fast local development and optimized builds.
React Router 7	Declarative routing engine for application navigation and guarded dashboard access.
i18next & react-i18next	Internationalization frameworks enabling multi-language translation support across the UI.
Lucide React	Clean, modern SVG icons for dashboard layouts, form buttons, and symptom tracking.
Node.js & Express.js	Server runtime and web framework for implementing secure REST APIs and routing middleware.
MongoDB & Mongoose	NoSQL database and ODM for storing data structures including users, specialists, and bookings.
Redis & ioredis	High-performance cache and in-memory datastore used to cache API responses and session metadata.
JSONWebToken & bcryptjs	Secure authentication tokens and password-hashing algorithms for role-based user access.
PDFKit	Server-side utility for compiling clinical triage summaries and generating downloadable receipts.
Multer	Middleware for handling multipart form data, facilitating symptom voice recording uploads.
Resend	Email dispatch API integration for multi-channel pre-visit and post-visit patient notifications.

System Architecture

The system architecture of the SmartAssist platform is structured around a decoupled full-stack paradigm to ensure maximum efficiency and modularity.
The user interface layer uses React 19 and Vite to deliver high-performance, responsive client dashboards and triage chat states.
The service layer is managed by a Node.js and Express.js backend that handles secure API routes, authentication workflows, and asynchronous integrations.
The data persistence layer combines a primary MongoDB Atlas database with a fast Redis caching engine to guarantee low-latency information retrieval.

System Architecture Design Flow Chart



Start
User
Input Symptoms
Voice?
Yes
No
Record Audio
Transcribe
Text
Text Processing
Analyze Triage
Recommendation
Specialist Found?
Yes
No
Booking
Search
Check Availability
Conflicting?
Yes
No
Join Waitlist
Reserve Slot
Slot Vacated?
No
Yes
Wait
Claim
Generate PDF Summary
Send Notification
End

Solution Overview

SmartAssist is a full-stack digital health platform linking patients with specialized care. Combining artificial intelligence with a live database engine, the platform functions as an automated triage nurse and scheduling assistant. It streamlines patient routing, eliminates administrative delays, and optimizes specialist resource utilization.

Patients enter symptom descriptions via text or audio recording. A server-side transcription daemon parses voice data before large language models analyze the input. The AI Triage Nurse classifies condition severity, provides urgency levels, issues emergency warnings, and recommends matching medical specialists.

Approved patients book appointments across three consultation modes: In-Person, Video, or Chat. The scheduling system prevents slot conflicts and runs a background waitlist module to reallocate cancelled slots. Guarded dashboards visualize patient symptom recovery and generate pre-visit summaries for medical practitioners.

Background worker processes automate scheduling cleanup operations and trigger SMS and email alerts at preset intervals. Standard security measures include JSON Web Token validation alongside salted password storage to prevent unauthorized profile access. Patient clinical receipts and intake diagnostics are converted to PDF summaries for prompt delivery. Deployment protocols leverage isolated environments to achieve optimal load balancing and high availability rates.

Functional and Non-Functional Requirements

Functional Requirements

- Interactive AI Triage Chat: The system must provide a conversational, multi-turn AI chat interface for symptom assessment.
- Audio Symptom Recording and Transcription: The application must capture patient audio symptom inputs and transcribe them on the server.
- Specialist Recommender System: The platform must analyze symptom severity and recommend matching medical specialist categories.
- Multi-Modal Appointment Booking: The scheduling engine must support In-Person, Video Call, and Chat bookings with real-time conflict check.
- Automated Waitlist Queue: The system must queue patients for booked slots and automatically reassign slots upon cancellation.
- PDF Summaries and Receipts: The application must generate clinical triage summaries and payment receipts as downloadable PDF files.
- Symptom Recovery Visualization: The system must display symptom progression over time via an interactive SVG chart on the user dashboard.
- Admin Approval Console: The application must provide administrators with interfaces to approve or reject pending specialist registrations.

Non-Functional Requirements

- Low-Latency Performance: Triage analysis responses and database booking actions must execute within two seconds under load.
- Robust Caching Efficiency: The platform must use Redis to cache specialist profile information and calendar availability.
- High Data Security: All user passwords must be hashed using bcrypt, and API routes must be protected using JWT verification.
- System Scalability: The system backend must scale dynamically to handle concurrent booking requests without double-booking slots.
- Mobile-Responsive Design: The user interface must adapt fluidly to desktop, tablet, and mobile displays with full responsiveness.

- High Availability Deployment: The production environment must maintain a 99.9% uptime rate for medical scheduling services.
- Multi-Language Adaptability: The user interface must dynamically translate text content without page reloads using i18n modules.
- Scalable Data Caching: The data access layer must fallback to local memory cache if connection to Redis fails.

Database Design

SmartAssist utilizes MongoDB Atlas as its primary NoSQL document database to manage complex, nested, and evolving medical records. By adopting MongoDB, the system can seamlessly store dynamic structures such as multi-turn triage chat transcripts and custom appointment configurations without rigid schema constraints. Additionally, an ioredis-based Redis caching service stores frequently requested availability slots and specialist profile summaries, reducing primary database load and accelerating response times.

User Collection Schema

Field Name	Data Type	Required	Purpose / Description
name	String	Yes	The user's full name
email	String	Yes	Unique, lowercase, trimmed email address for authentication
password	String	Yes	Bcrypt hashed password string
role	String	No	User role: 'user', 'specialist', or 'admin' (default: 'user')
familyProfiles	Array	No	Nested sub-documents containing names and relationships of family members
waitlistAppointments	Array	No	Nested sub-documents tracking queued bookings and claims
notificationPreferences	Object	No	SMS and Email alerts enablement toggle (default: true)

Specialist Collection Schema

Field Name	Data Type	Required	Purpose / Description
userId	ObjectId	No	Reference to the corresponding User collection document
name	String	Yes	Specialist's professional name
specialization	String	Yes	Area of medical or wellness expertise
experience	String	No	Years or summary of professional experience
bio	String	No	Professional bio and description
rating	Number	No	Average user rating (default: 5)
licenseNumber	String	No	Unique license registration number for admin approval
status	String	No	Verification state: 'pending', 'approved', 'rejected'
rejectionReason	String	No	Reason if the specialist registration application was rejected
clinicName	String	No	Name of the associated physical clinic
address	String	No	Physical address of the clinic location
appointmentModes	Object	No	Details (enabled, pricing, slots) for In-Person, Video, and Chat modes

Booking Collection Schema

Field Name	Data Type	Required	Purpose / Description
receiptId	String	Yes	Unique billing and tracking receipt ID
userName	String	Yes	Name of the patient booking the slot
userEmail	String	Yes	Email of the booking user
specialistId	String	Yes	Target specialist ID identifier
bookingDate	String	Yes	Date of the scheduled appointment
bookingTime	String	Yes	Reserved time slot of the appointment
status	String	No	Appointment state (e.g. 'confirmed', 'cancelled')
appointmentMode	String	No	Consultation channel: 'In-Person', 'Video', or 'Chat'
price	Number	No	Booking price paid for the consultation
duration	String	No	Duration of the appointment slot
triageUrgency	String	No	Urgency level determined by AI triage nurse
triageExplanation	String	No	Clinical justification generated by AI model
triageHistory	Array	No	Nested sub-documents of triage conversational transcripts
postVisitFeedback	Object	No	Patient recovery updates, flagged indicators, and follow-ups

API Integration

Resend API

- Transmits automated pre-appointment reminders to patients at 24-hour, 2-hour, and 15-minute intervals.
- Dispatches post-visit follow-up emails for rating specialist consultations and tracking recovery symptoms.
- Delivers automated email alerts regarding specialist registration onboarding, approval, or rejection status.
- Generates and attaches clinical pre-visit summary PDFs directly to specialists one hour prior to consultation.

Gemini API

- Acts as the primary conversational artificial intelligence model (gemini-3.1-flash-lite) for patient symptom intake.
- Evaluates multi-turn chat interactions to extract key health concerns and match confidence scores.
- Classifies user symptoms into predefined specialist categories including Dentist, Physiotherapist, Gym Trainer, and Salon Specialist.
- Generates urgency assessment levels (Urgent, Soon, Routine) and formats structured clinical reports for doctors.

OpenAI API

- Operates as a secondary AI provider utilizing the gpt-4o-mini engine for structured triage analysis.
- Parses complex symptom descriptions to identify medical keywords and evaluate patient risk factors.
- Formulates dynamic fallback recommendations when primary AI services experience high traffic or timeouts.
- Sanitizes and formats raw patient symptom logs into structured JSON payloads for backend processing.

Groq API

- Integrates the llama-3.3-70b-versatile model to deliver ultra-low latency AI triage responses.
- Serves as a high-speed fallback provider ensuring continuous availability during peak system loads.
- Performs rapid keyword extraction and sentiment analysis on patient voice recording transcripts.

- Validates specialist category matching against historical triage datasets with high inference speeds.

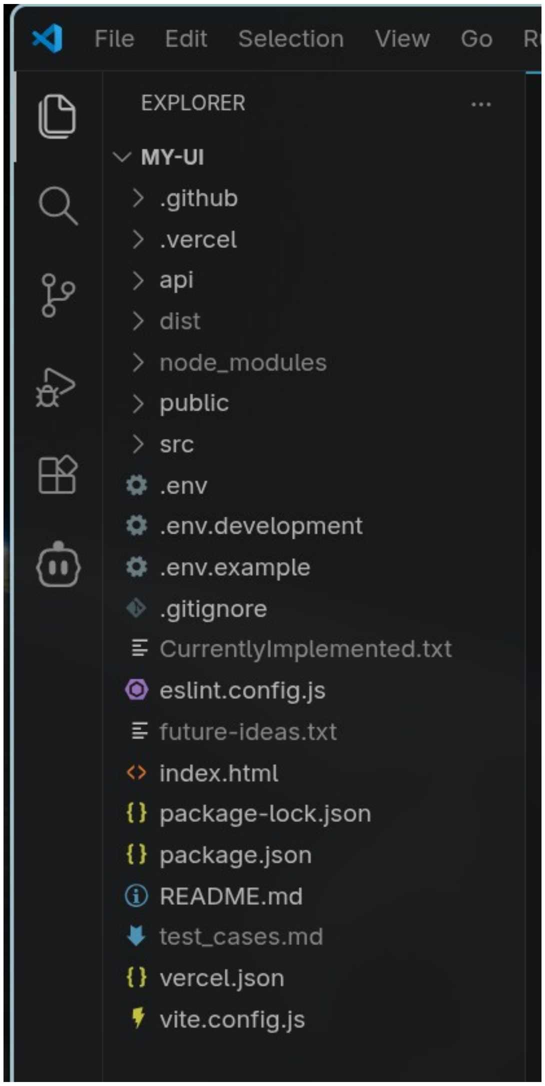
Google Maps API

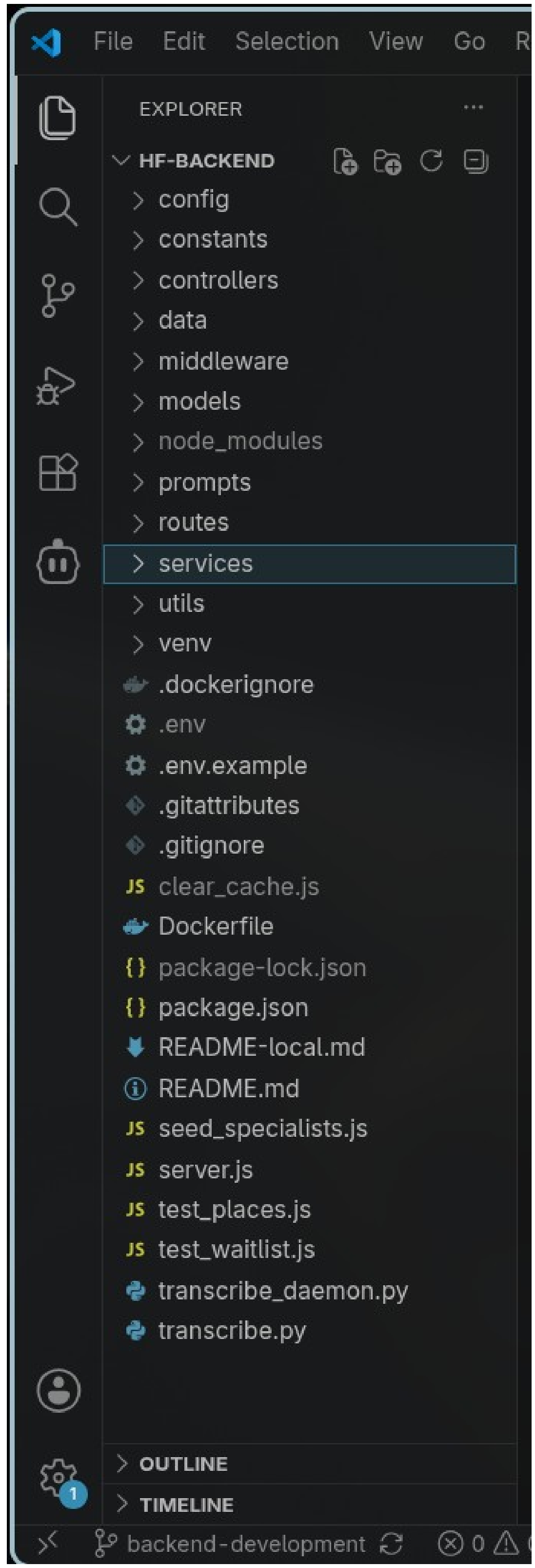
- Integrates Google Places service to fetch verified geographic locations for medical and wellness specialists.
- Calculates proximity distances between patient zip codes and physical clinic addresses for in-person bookings.
- Provides geocoding capabilities to convert clinic street addresses into interactive map coordinates.
- Displays dynamic map previews on specialist profile cards to assist patients with navigation.

Textracts

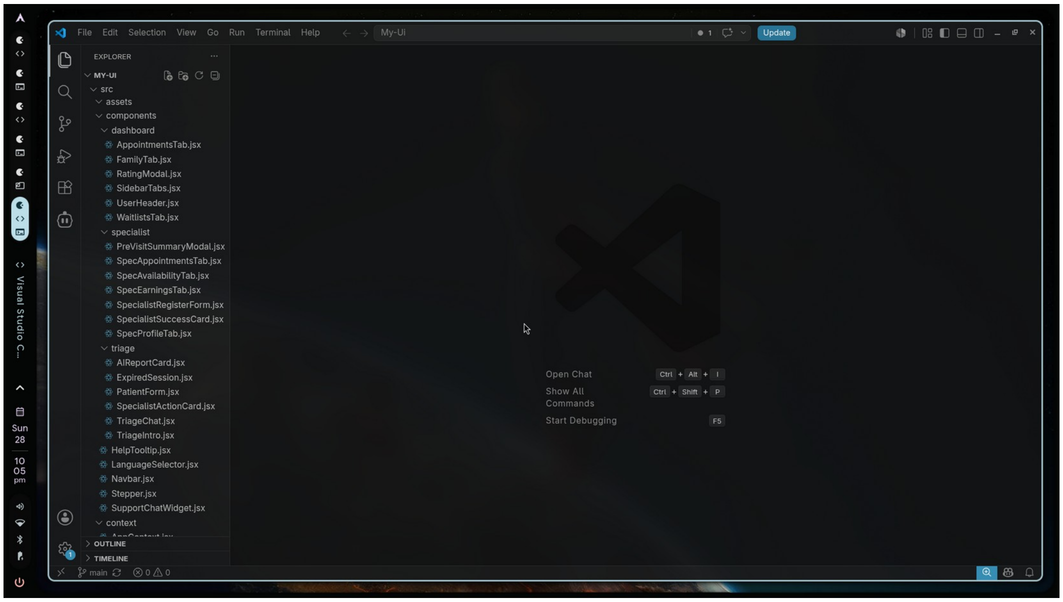
The following technical extracts demonstrate the core operational architecture, project file structure, frontend user interface implementation, and backend API service configurations powering the SmartAssist platform.

Project Folder Structure

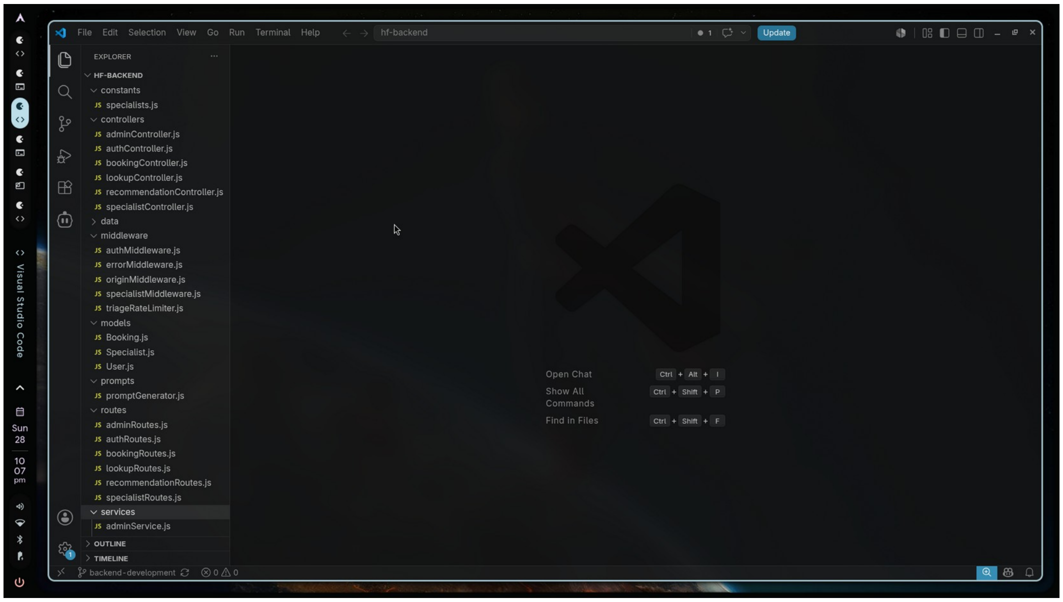




Frontend



Backend



Testing and Validation

Testing and validation were conducted to ensure the reliability, security, performance, and functional accuracy of the SmartAssist platform.

Functional Testing

- Verified AI triage conversation flows, symptom parsing, and specialist category recommendations.
- Checked real-time calendar slot booking, patient detail inputs, and receipt ticket generation.
- Validated specialist dashboard profile updates, slot availability toggles, and mode pricing configurations.

Integration Testing

- Tested seamless data synchronization between React frontend services and Node/Express backend APIs.
- Verified MongoDB database read/write queries for user accounts, bookings, and waitlist state persistence.
- Tested external API integrations including Gemini, OpenAI, Groq, Resend email service, and Google Maps.

Validation Testing

- Conducted user authentication security checks, verifying password bcrypt hashing and JWT token authorization.
- Evaluated automated background cron jobs for pre-appointment reminders and waitlist slot reallocation.
- Tested responsive UI rendering and accessibility cross-browser performance on desktop and mobile devices.

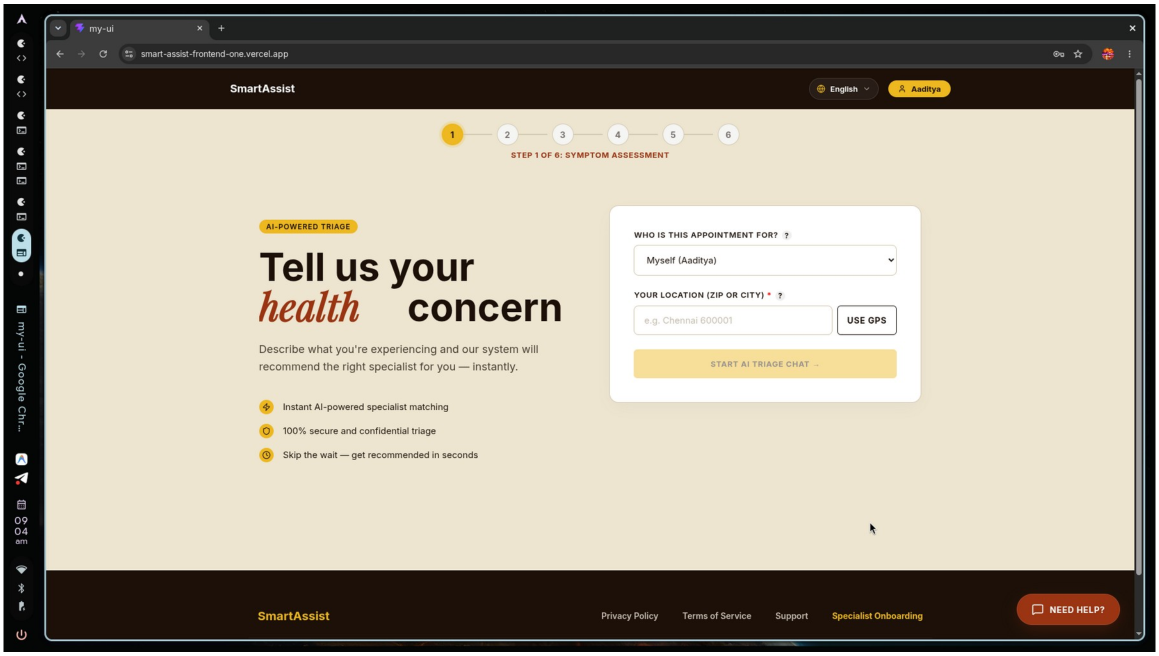
Sample Test Cases

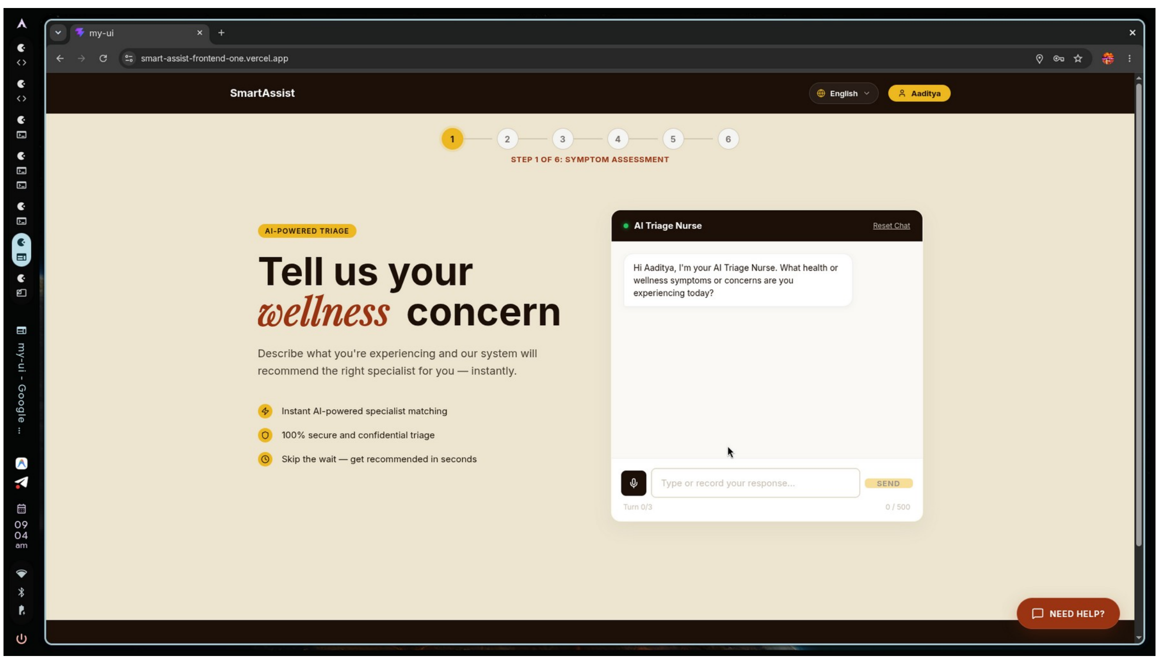
Test Case	Status
Verify AI triage symptom classification and specialist recommendation	**PASSED**
Test user registration and JWT authentication token issuance	**PASSED**
Validate appointment slot booking and occupied slot conflict detection	**PASSED**
Verify automated waitlist queue joining and hold expiration handling	**PASSED**
Test automated email reminder dispatch via Resend API integration	**PASSED**
Verify pre-visit clinical summary PDF generation for specialists	**PASSED**

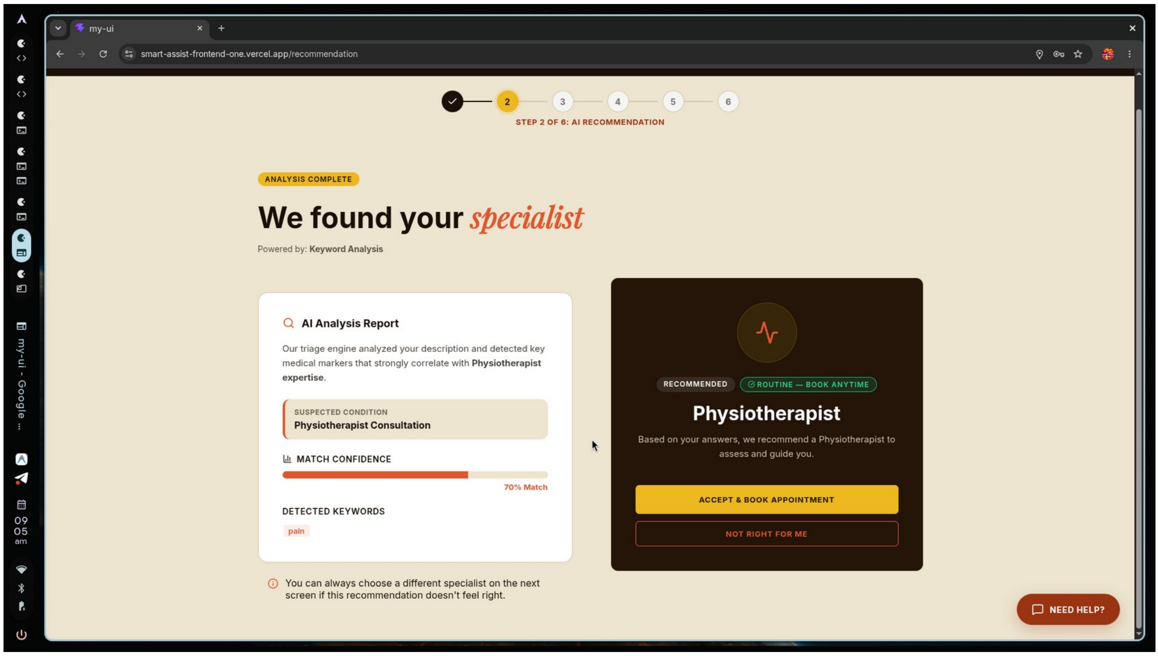
Result

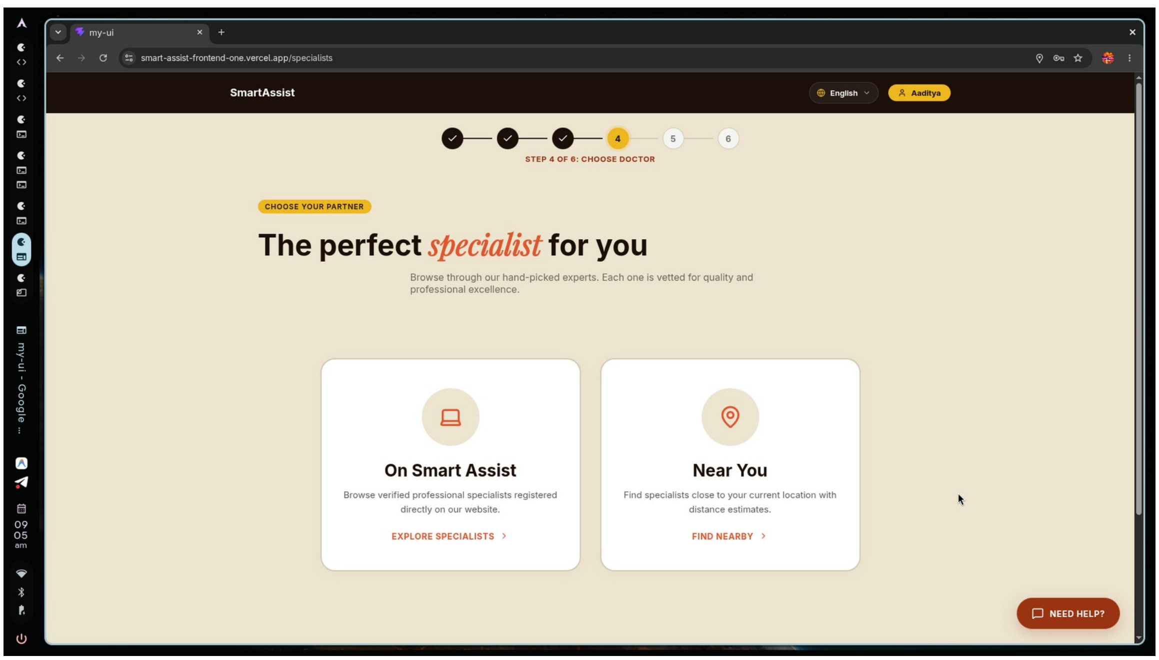
The system successfully met all functional requirements, security benchmarks, and operational standards during testing and validation.

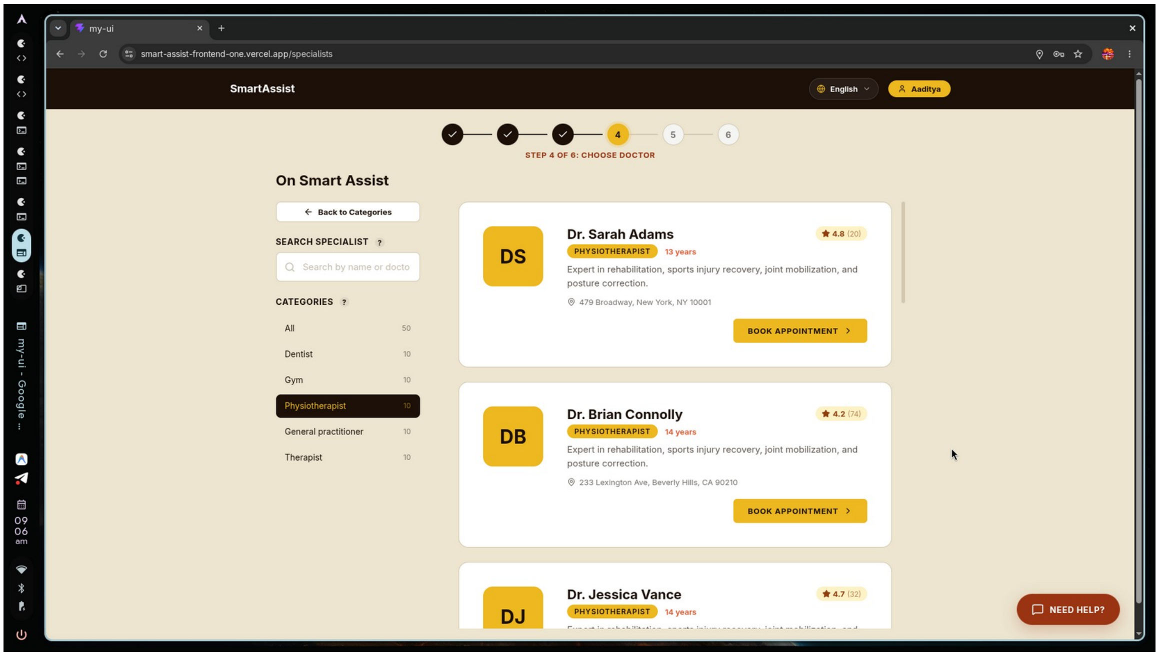
Customer Journey






my-ui
smart-assist-frontend-one.vercel.app/recommendation
STEP 2 OF 6: AI RECOMMENDATION
ANALYSIS COMPLETE
We found your specialist
Powered by: Keyword Analysis
AI Analysis Report
Our triage engine analyzed your description and detected key medical markers that strongly correlate with Physiotherapist expertise.
SUSPECTED CONDITION
Physiotherapist Consultation
MATCH CONFIDENCE
70% Match
DETECTED KEYWORDS
pain
RECOMMENDED
ROUTINE — BOOK ANYTIME
Physiotherapist
Based on your answers, we recommend a Physiotherapist to assess and guide you.
ACCEPT & BOOK APPOINTMENT
NOT RIGHT FOR ME
You can always choose a different specialist on the next screen if this recommendation doesn't feel right.
NEED HELP?


my-ui
smart-assist-frontend-one.vercel.app/specialists
SmartAssist
English
Aaditya
STEP 4 OF 6: CHOOSE DOCTOR
CHOOSE YOUR PARTNER
The perfect specialist for you
Browse through our hand-picked experts. Each one is vetted for quality and professional excellence.
On Smart Assist
Browse verified professional specialists registered directly on our website.
EXPLORE SPECIALISTS
Near You
Find specialists close to your current location with distance estimates.
FIND NEARBY
NEED HELP?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English
Aaditya
STEP 4 OF 6: CHOOSE DOCTOR
On Smart Assist
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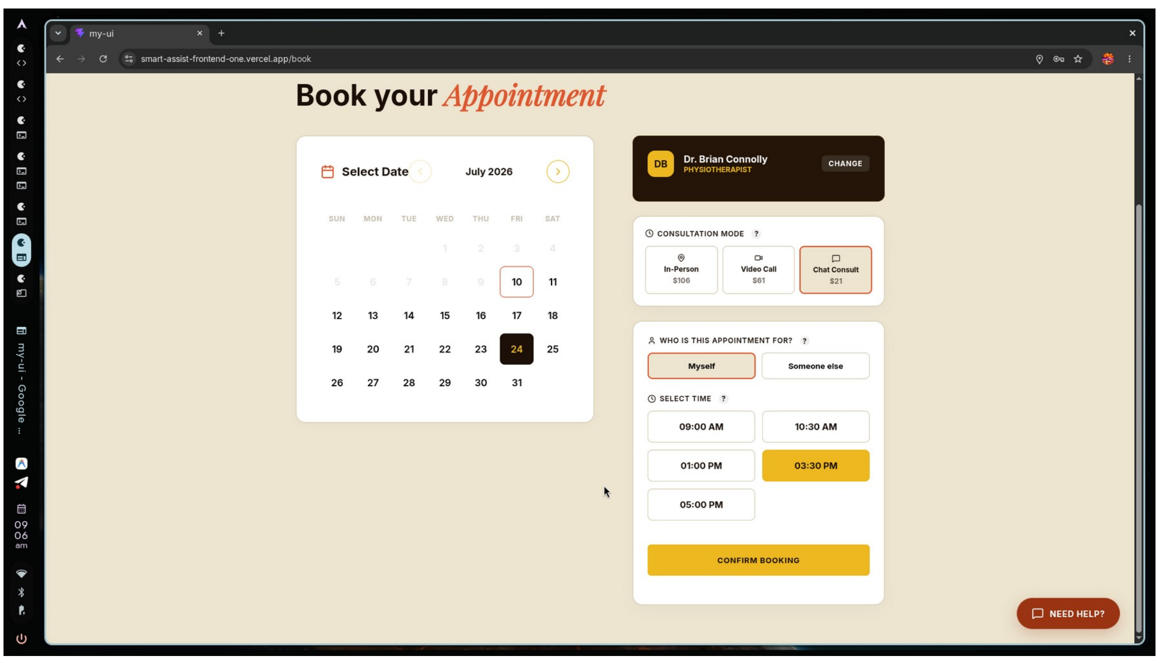
SEARCH SPECIALIST
Search by name or docto
CATEGORIES
All 50
Dentist 10
Gym 10
Physiotherapist 10
General practitioner 10
Therapist 10
DS
Dr. Sarah Adams
4.8 (20)
PHYSIOTHERAPIST
13 years
Expert in rehabilitation, sports injury recovery, joint mobilization, and posture correction.
479 Broadway, New York, NY 10001
BOOK APPOINTMENT
DB
Dr. Brian Connolly
4.2 (74)
PHYSIOTHERAPIST
14 years
Expert in rehabilitation, sports injury recovery, joint mobilization, and posture correction.
233 Lexington Ave, Beverly Hills, CA 90210
BOOK APPOINTMENT
DJ
Dr. Jessica Vance
4.7 (32)
PHYSIOTHERAPIST
14 years
NEED HELP?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Book your Appointment
Select Date
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5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
DB
Dr. Brian Connolly
PHYSIOTHERAPIST
CHANGE
CONSULTATION MODE
In-Person $106
Video Call $61
Chat Consult $21
WHO IS THIS APPOINTMENT FOR?
Myself
Someone else
SELECT TIME
09:00 AM
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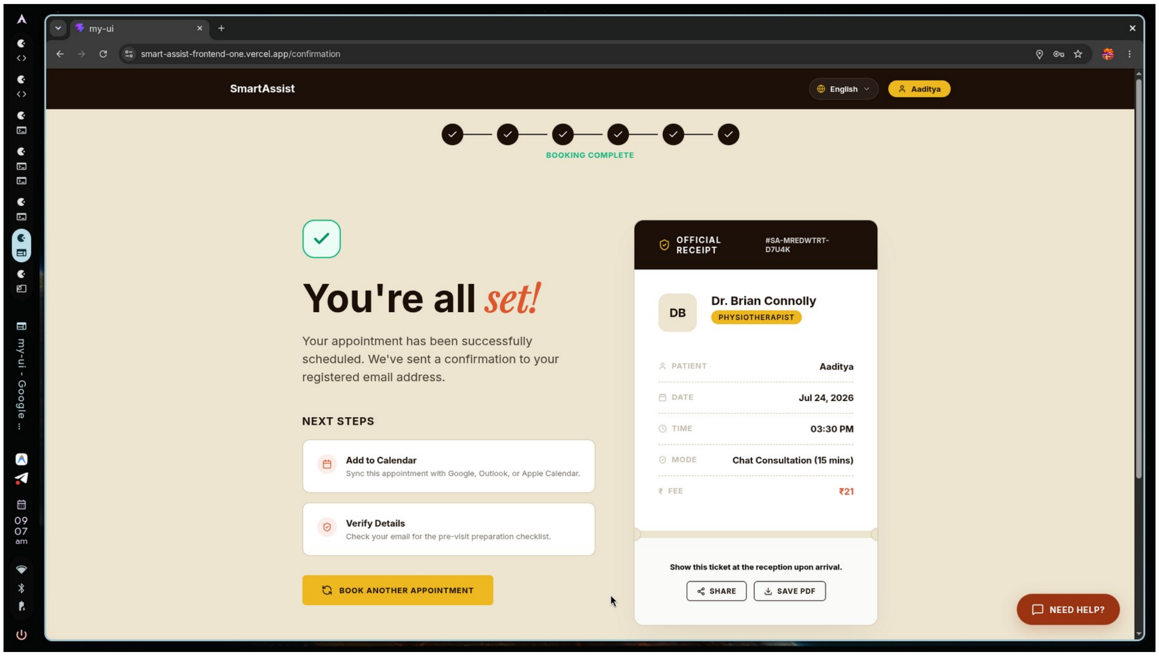
05:00 PM
CONFIRM BOOKING
NEED HELP?


my-ui
smart-assist-frontend-one.vercel.app/confirmation
SmartAssist
English
Aaditya
BOOKING COMPLETE
You're all set!
Your appointment has been successfully scheduled. We've sent a confirmation to your registered email address.
NEXT STEPS
Add to Calendar
Sync this appointment with Google, Outlook, or Apple Calendar.
Verify Details
Check your email for the pre-visit preparation checklist.
BOOK ANOTHER APPOINTMENT
OFFICIAL RECEIPT
#SA-MREDWTRT-D7U4K
DB
Dr. Brian Connolly
PHYSIOTHERAPIST
PATIENT
Aaditya
DATE
Jul 24, 2026
TIME
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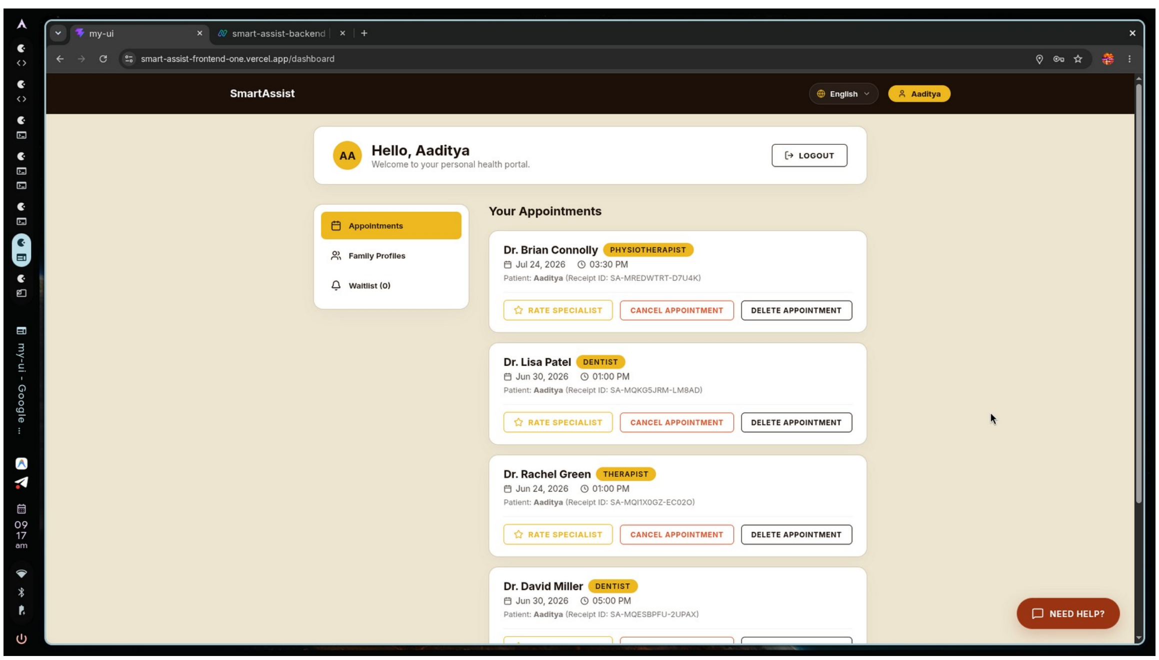
MODE
Chat Consultation (15 mins)
FEE
₹21
Show this ticket at the reception upon arrival.
SHARE
SAVE PDF
NEED HELP?


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/dashboard
SmartAssist
English
Aaditya
AA
Hello, Aaditya
Welcome to your personal health portal.
LOGOUT
Appointments
Family Profiles
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Your Appointments
Dr. Brian Connolly
PHYSIOTHERAPIST
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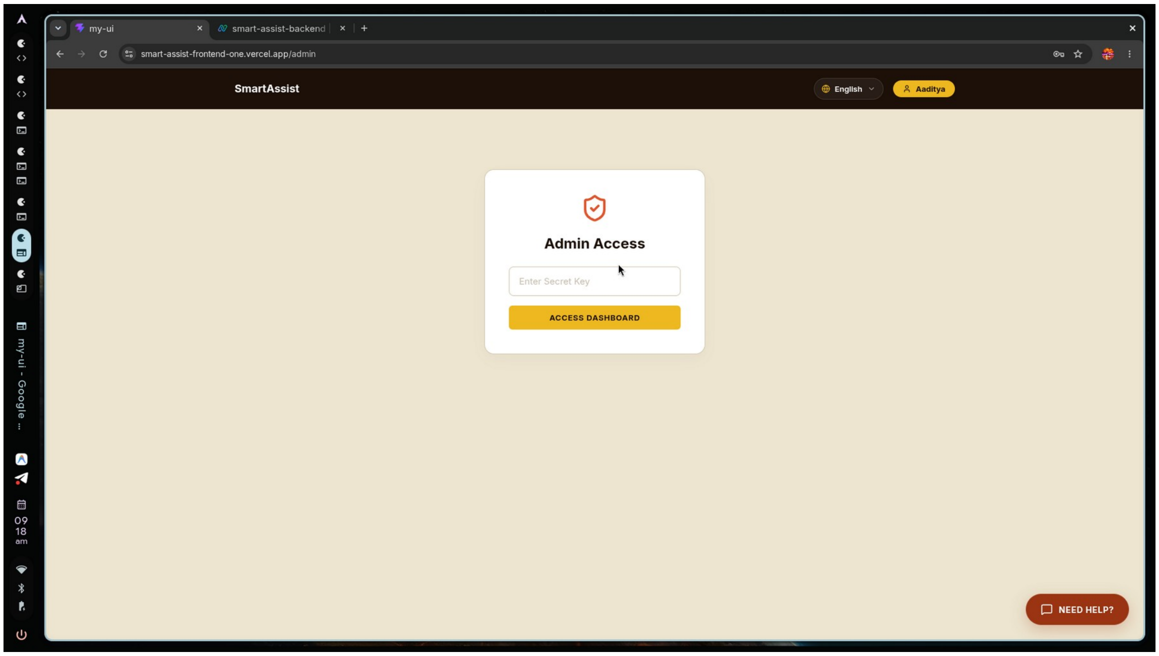
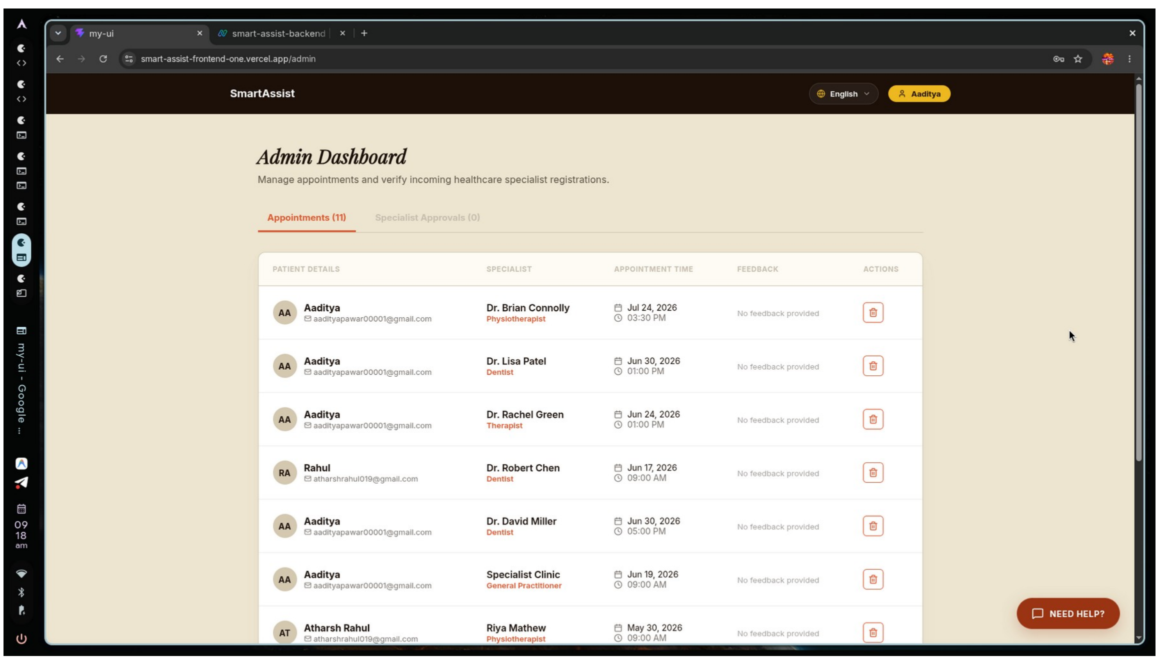
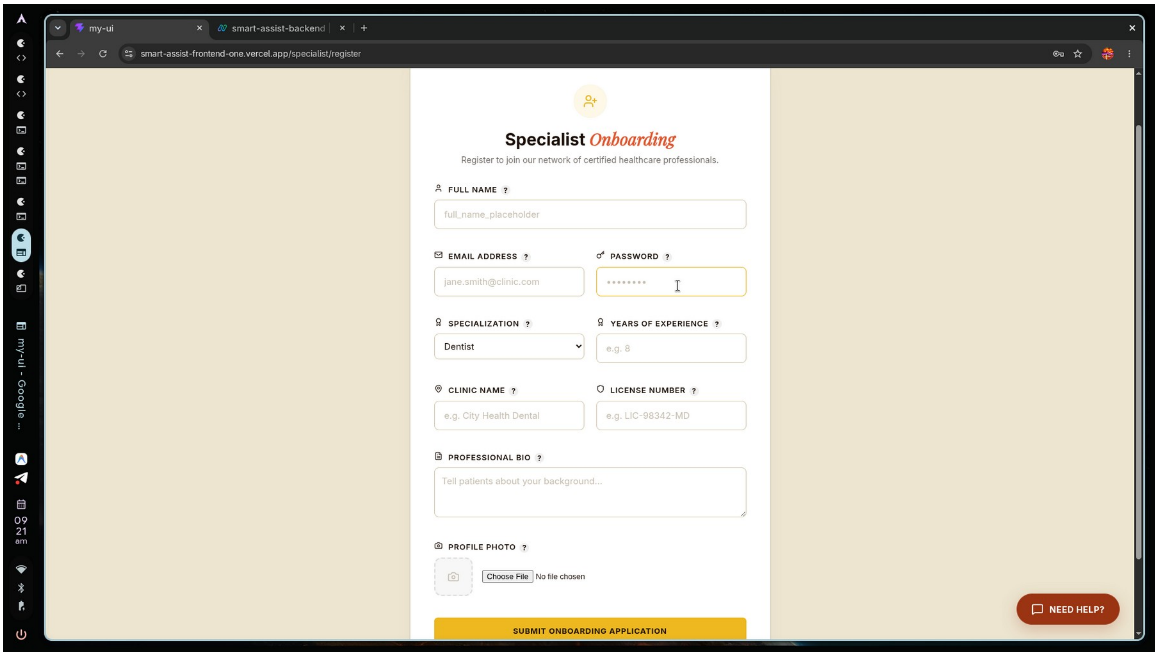
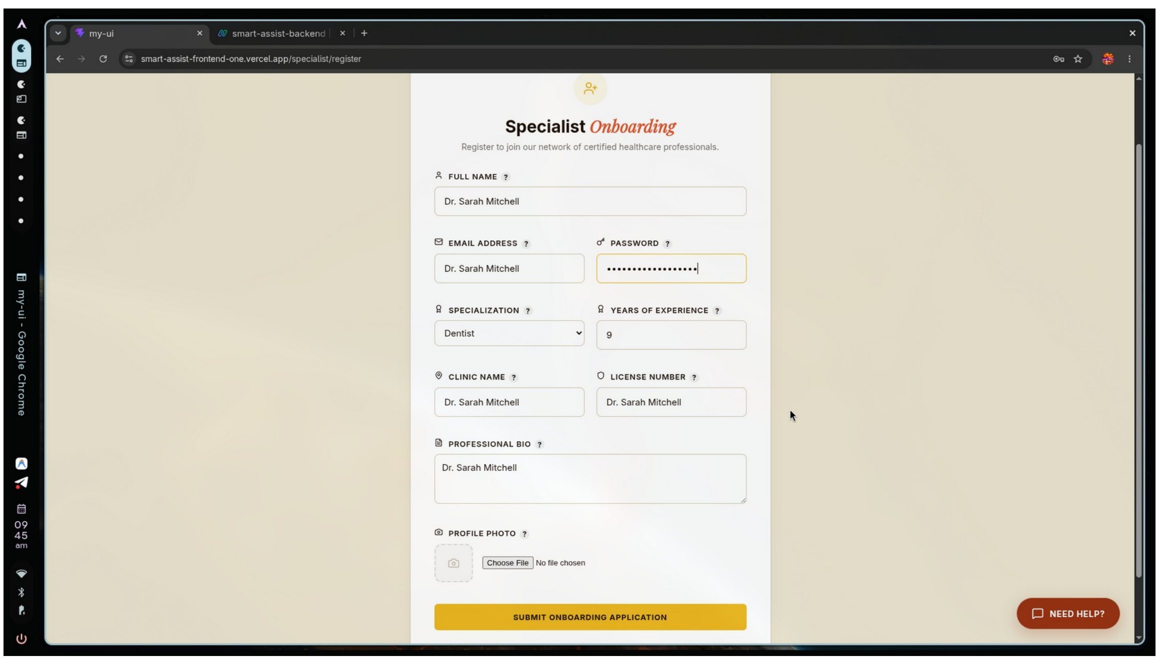
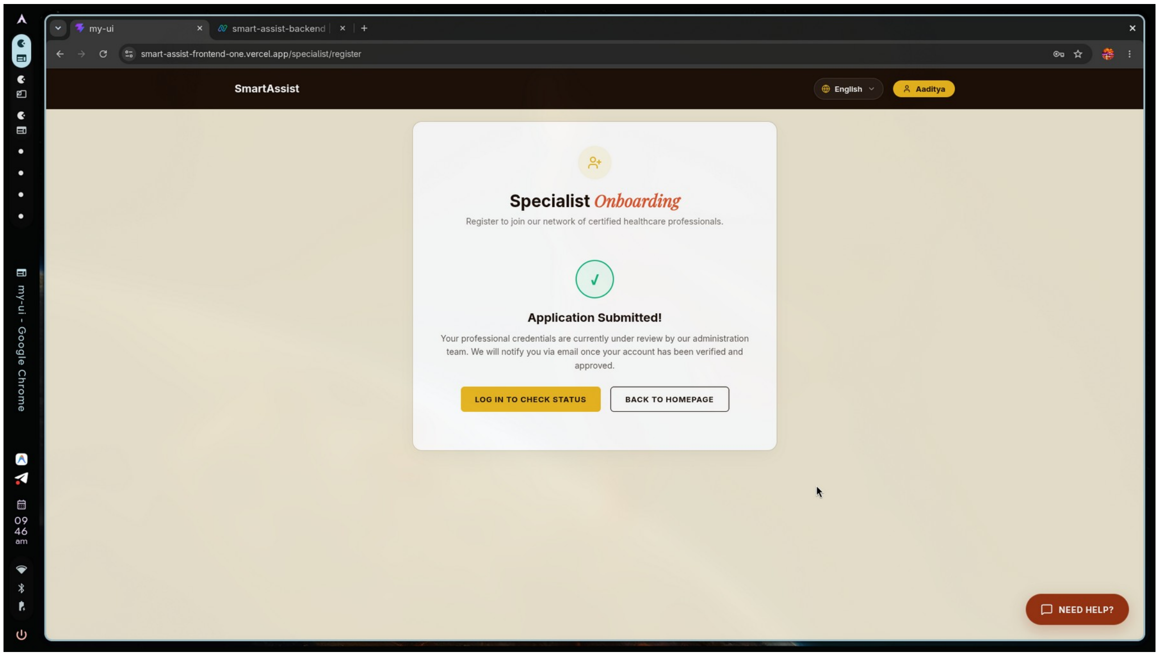
NEED HELP?

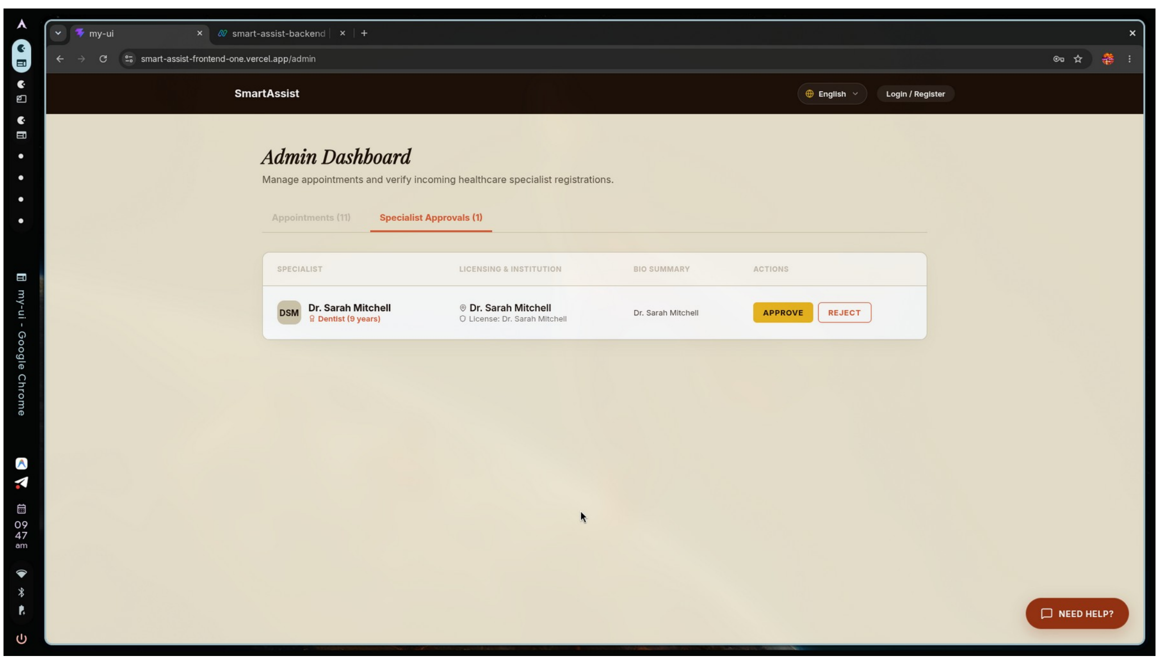

my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/admin
SmartAssist
English
Aaditya
Admin Access
Enter Secret Key
ACCESS DASHBOARD
NEED HELP?
09
18
am
my-ui - Google ...

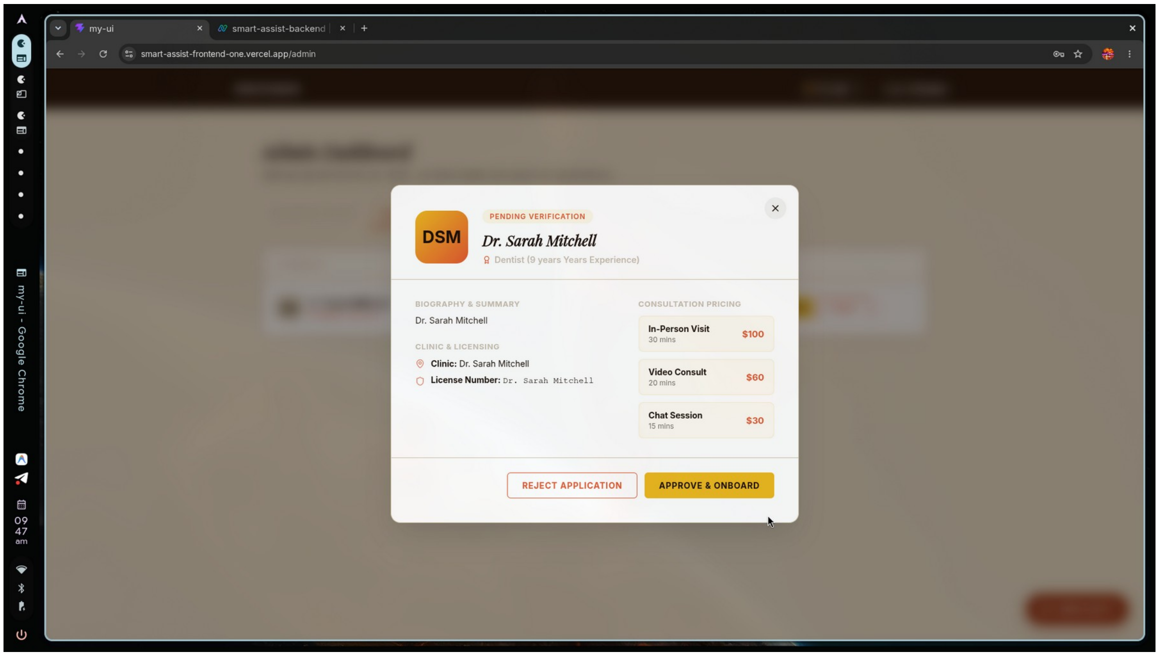

my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/admin
SmartAssist
English
Aaditya
Admin Dashboard
Manage appointments and verify incoming healthcare specialist registrations.
Appointments (11)
Specialist Approvals (0)
PATIENT DETAILS
SPECIALIST
APPOINTMENT TIME
FEEDBACK
ACTIONS
AA Aaditya aadityapawar00001@gmail.com
Dr. Brian Connolly Physiotherapist
Jul 24, 2026 03:30 PM
No feedback provided
AA Aaditya aadityapawar00001@gmail.com
Dr. Lisa Patel Dentist
Jun 30, 2026 01:00 PM
No feedback provided
AA Aaditya aadityapawar00001@gmail.com
Dr. Rachel Green Therapist
Jun 24, 2026 01:00 PM
No feedback provided
RA Rahul atharshrahul019@gmail.com
Dr. Robert Chen Dentist
Jun 17, 2026 09:00 AM
No feedback provided
AA Aaditya aadityapawar00001@gmail.com
Dr. David Miller Dentist
Jun 30, 2026 05:00 PM
No feedback provided
AA Aaditya aadityapawar00001@gmail.com
Specialist Clinic General Practitioner
Jun 19, 2026 09:00 AM
No feedback provided
AT Atharsh Rahul atharshrahul019@gmail.com
Riya Mathew Physiotherapist
May 30, 2026 09:00 AM
No feedback provided
NEED HELP?
09
18
am
my-ui - Google ...


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/specialist/register
Specialist Onboarding
Register to join our network of certified healthcare professionals.
FULL NAME
full_name_placeholder
EMAIL ADDRESS
PASSWORD
jane.smith@clinic.com
SPECIALIZATION
YEARS OF EXPERIENCE
Dentist
e.g. 8
CLINIC NAME
LICENSE NUMBER
e.g. City Health Dental
e.g. LIC-98342-MD
PROFESSIONAL BIO
Tell patients about your background...
PROFILE PHOTO
Choose File No file chosen
SUBMIT ONBOARDING APPLICATION
NEED HELP?
09 21 am


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/specialist/register
Specialist Onboarding
Register to join our network of certified healthcare professionals.
FULL NAME
Dr. Sarah Mitchell
EMAIL ADDRESS
PASSWORD
Dr. Sarah Mitchell
SPECIALIZATION
YEARS OF EXPERIENCE
Dentist
9
CLINIC NAME
LICENSE NUMBER
Dr. Sarah Mitchell
Dr. Sarah Mitchell
PROFESSIONAL BIO
Dr. Sarah Mitchell
PROFILE PHOTO
Choose File No file chosen
SUBMIT ONBOARDING APPLICATION
NEED HELP?
09 45 am


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/specialist/register
SmartAssist
English
Aaditya
Specialist Onboarding
Register to join our network of certified healthcare professionals.
Application Submitted!
Your professional credentials are currently under review by our administration team. We will notify you via email once your account has been verified and approved.
LOG IN TO CHECK STATUS
BACK TO HOMEPAGE
NEED HELP?


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/admin
SmartAssist
English
Login / Register
Admin Dashboard
Manage appointments and verify incoming healthcare specialist registrations.
Appointments (11)
Specialist Approvals (1)
SPECIALIST
LICENSING & INSTITUTION
BIO SUMMARY
ACTIONS
DSM
Dr. Sarah Mitchell
Dentist (9 years)
Dr. Sarah Mitchell
License: Dr. Sarah Mitchell
Dr. Sarah Mitchell
APPROVE
REJECT
NEED HELP?


my-ui
smart-assist-backend
smart-assist-frontend-one.vercel.app/admin
PENDING VERIFICATION
DSM
Dr. Sarah Mitchell
Dentist (9 years Years Experience)
BIOGRAPHY & SUMMARY
Dr. Sarah Mitchell
CLINIC & LICENSING
Clinic: Dr. Sarah Mitchell
License Number: Dr. Sarah Mitchell
CONSULTATION PRICING
In-Person Visit
30 mins
$100
Video Consult
20 mins
$60
Chat Session
15 mins
$30
REJECT APPLICATION
APPROVE & ONBOARD
my-ui - Google Chrome
09
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am

Project Outcome

1. Automated Clinical Triage Efficiency

- Reduction in patient wait times by automating symptom assessment and initial screening.
- Elimination of human error in routing by classifying cases into clear urgency tiers.

2. Direct and Accurate Specialist Matching

- Seamless routing of patients to the correct department based on conversational AI analysis.
- Decreased rate of misrouted appointments and incorrect clinic bookings.

3. Optimized Slot Utilization and Scheduling

- Automated reallocation of cancelled slots, ensuring clinic hours are fully utilized.
- Implementation of a real-time waitlist that instantly fills open gaps.

4. Comprehensive Pre-Visit Context for Doctors

- Automatic generation of detailed symptom summaries before the consultation starts.
- Improved clinical efficiency as intake data is collected and formatted beforehand.

5. High Patient Engagement and Accessibility

- Conversational voice-and-text interface allowing easy interaction for all users.
- Instant digital receipts and instructions sent immediately upon booking confirmation.

6. Overall Impact

- Significant transformation of healthcare operations through automated intake workflows.
- Marked improvement in clinician utilization and overall patient satisfaction ratings.

Future Enhancements

1. AI Health Tips and Suggestions

- Deliver personalized preventive health recommendations based on patient triage history.
- Provide daily or weekly wellness tips tailored to individual risk factors.

2. Integrated Telehealth Consultations

- Support one-click video and audio appointments directly inside the web interface.
- Implement real-time screen sharing and chat capabilities during virtual consultations.

3. Wearable Device Integration

- Sync real-time vital data like heart rate and sleep patterns from smartwatches.
- Trigger automated alerts to doctors if patient vitals cross safe thresholds.

4. Electronic Health Record (EHR) Syncing

- Automatically push generated clinical summaries directly to global hospital databases.
- Pull historical medical records to offer deeper context during AI triage.

5. Prescriptions and Pharmacy Routing

- Enable doctors to send digital prescriptions to local pharmacies instantly.
- Let patients select their preferred pickup locations and check medication availability.

6. Post-Consultation Follow-ups

- Schedule automated check-ins via AI chat to monitor recovery progress.
- Prompt patients to report side effects or schedule follow-up appointments when needed.

Conclusion

The development and implementation of the SmartAssist platform represents a significant advancement in bridging the gap between artificial intelligence and day-to-day clinical operations. By transitioning from traditional, labor-intensive triage methodologies to a conversational, voice-enabled AI interface, the project successfully demonstrates how modern technologies can reduce patient onboarding bottlenecks. The automation of initial symptom classification ensures that individuals seeking care are routed to the appropriate medical specialist with high accuracy, minimizing misallocated slots and reducing initial diagnostic wait times.

From a technical perspective, the platform integrates several sophisticated layers: automated speech-to-text transcription, real-time classification engines, dynamic database models, and smart waitlists. By enabling real-time appointment booking and automating slot reallocation upon cancellation, the system addresses the persistent challenge of clinic underutilization. This seamless coordination between patient inputs, database states, and live doctor schedules creates an end-to-end medical assistant tool that operates reliably under fluctuating workloads.

Operational efficiency is further enhanced by the automatic generation of structured clinical summaries. Providing doctors with comprehensive triage history and symptom context prior to the start of a consultation allows healthcare professionals to optimize session times. Rather than spending valuable diagnostic minutes on basic demographic and intake questionnaires, clinicians can immediately address active patient needs, boosting patient satisfaction and enabling clinics to manage higher patient volumes safely.

Looking forward, the successful deployment of SmartAssist sets a strong foundation for future integrations. By incorporating wearable vital monitoring, multi- EHR synchronization, automated electronic prescriptions, and personalized health recommendations, the platform is well-positioned to evolve into a comprehensive digital health ecosystem. Ultimately, SmartAssist proves that intelligent automation is not only viable for administrative support but is an essential component of modern, patient-centric healthcare systems.